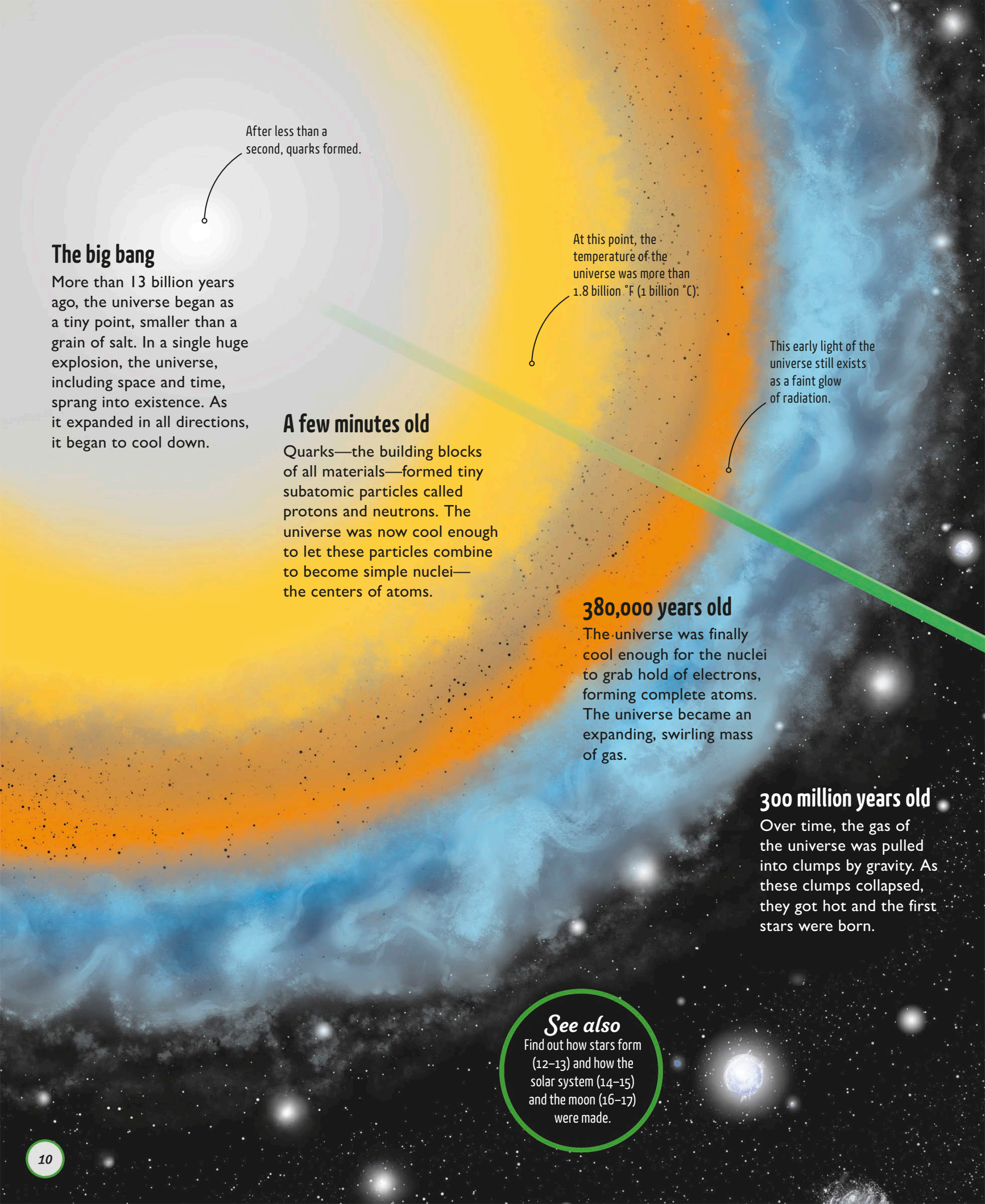




After less than a second, quarks formed.

The big bang

More than 13 billion years ago, the universe began as a tiny point, smaller than a grain of salt. In a single huge explosion, the universe, including space and time, sprang into existence. As it expanded in all directions, it began to cool down.

At this point, the temperature of the universe was more than 1.8 billion °F (1 billion °C).

This early light of the universe still exists as a faint glow of radiation.

A few minutes old

Quarks—the building blocks of all materials—formed tiny subatomic particles called protons and neutrons. The universe was now cool enough to let these particles combine to become simple nuclei—the centers of atoms.

380,000 years old

The universe was finally cool enough for the nuclei to grab hold of electrons, forming complete atoms. The universe became an expanding, swirling mass of gas.

300 million years old

Over time, the gas of the universe was pulled into clumps by gravity. As these clumps collapsed, they got hot and the first stars were born.

See also

Find out how stars form (12–13) and how the solar system (14–15) and the moon (16–17) were made.